

Isaias Perez

isaiasxl21@gmail.com | [linkedin.com/in/isaias-perez21](https://www.linkedin.com/in/isaias-perez21) | isaias-perez-portfolio.onrender.com | (720) 980-7488

SUMMARY

Software engineer and forward deployment intern graduating **May 2027**, available **June 2027**. Ships customer-facing AI systems across **retrieval, agent evaluation, data pipelines, and performance-critical software**, with production experience at Apple and deployment work across enterprise-search and restaurant pilots.

EXPERIENCE

Forward Deployment Engineer Intern

Sep 2026 – Present

Sinequa – ChapsVision

Remote

- Built and deployed a hosted enterprise-search POC for an **aerospace and defense manufacturer**, integrating multimodal ingestion, neural search, and a grounded AI assistant into a customer-ready technical demo.
- Translated discovery requirements into **source-level access controls, metadata filters, inline citations, document previews, and evidence-bounded answers**, enabling technical evaluators to verify every response against the underlying files.
- Conducted workflow discovery across sales, presales, and delivery and authored the architecture for reusable **agentic POC sandboxes** designed to reduce one-off demo work and strengthen opportunity-to-delivery handoffs.

Software Engineering Intern

May 2026 – Aug 2026

Apple – Apple TV / Services Engineering

Seattle, WA

- Cut AI-agent recovery latency by **~85%** and manual intervention by **~70%** by building event-driven **Swift/C** developer tooling with real-time state propagation.
- Built a self-healing Swift recovery command and **async-signal-safe C** crash sentinel that surfaced **SIGABRT/SIGSEGV** faults as structured diagnostics in **~1s**, down from **45s+**.
- Improved engineering-guidance reliability by **~40%** across **447 traceable intents in 9 domains** with confidence gates, provenance, conflict detection, and human ratification.
- Shipped customer-facing **Apple TV shelf updates** to production and debugged integration issues across a mature, cross-functional tvOS codebase.

Co-Founder and Lead Engineer

Jan 2026 – May 2026

CulinaryOS | Agentic Operational Intelligence Platform

Boulder, CO

- Led a **five-engineer team** building an operational intelligence platform across **4 pilot restaurants, 16 integrations, and 40+ asynchronous jobs**; fine-tuned and deployed a model to one pilot within **48 hours**.
- Architected fault-tolerant ingestion and orchestration with **PostgreSQL lease locks, idempotent writes, heartbeats, and supervisor recovery** across POS, labor, reservation, inventory, and service telemetry.
- Built a real-time **Operational State engine and agent decision layer** that converts live restaurant data into staffing, table-flow, ordering, and service recommendations with manager feedback and outcome tracking.
- Reduced held-out forecasting error by **61% (37.0% to 14.3% MAPE)** through leakage-safe temporal validation, site calibration, and gradient-boosting and Temporal Fusion Transformer models.

AI Engineer Intern

Jul 2025 – May 2026

Kroenke Sports and Entertainment

Denver, CO

- Built Python/PySpark/Databricks/Azure ML pipelines across **190K+ logs**, converting **232 PCI rows** into **188 actionable issues** and **169 remediation targets**.

Machine Learning Researcher

Mar 2025 – May 2026

University of Colorado Boulder – DLA and FUTURE Programs

Boulder, CO

- Cut RL experiment runtime by **33%** by profiling batched **PyTorch/CUDA** execution across **500K+ simulations**; improved evaluation reliability by **25%** with Redis-cached monitoring.
- Reduced LLM hallucinations by **15%** through attribution, saliency, and error analysis across transformer models while contributing explainability evaluations to an **ICML submission**.

PROJECTS

FlowExec++ | GPU-Accelerated RL Execution Simulator

C++, CUDA, PyTorch, Linux

- Increased replay and RL simulation throughput by **10x** and reduced simulated slippage by **7%** by engineering and profiling a C++/CUDA market simulator with stochastic impact models.

AdvisorGraph | Governed AI-Agent Evaluation

Python, PyTorch, Agentic Workflows

- Intercepted **100% of 1,666 injected violations** across **2,500 synthetic workflows** with a planner–retriever–policy–critic pipeline, confidence gates, citation checks, and reproducible audit records.

TECHNICAL SKILLS

Systems: C/C++, CUDA, Swift, Objective-C, Linux/UNIX, concurrency, profiling, crash and fault analysis

AI and Data: Python, PyTorch, SQL, RAG, agent workflows, evaluation, vector search, MCP, forecasting

Cloud and Product: TypeScript, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, Azure ML, Databricks, PySpark

EDUCATION

University of Colorado Boulder

Major GPA: 3.78

B.S. Computer Science; Minors in Applied Mathematics and Business

Expected May 2027

Selected coursework: Operating Systems, Computer Architecture, AI, ML, Databases, Algorithms, Probability, Linear Algebra